

Not Present

TMM WRITTEN TEST

- What is a stacking fault lattice defect?

(a) typical of BCC crystals	☒ typical of FCC crystals	(c) always present in metals	(d) typical of HCP crystals
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- Crystalline anisotropy affects mechanical properties in:

(a) HCP crystal	(b) FCC crystal	(c) BCC crystal	☒ all crystals
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Legend of Symbols⁽⁷⁾:
 TM = tempered martensite, B = bainite
 BL = lower bainite, P = pearlite
 AR = residual austenite, F = ferrite,
 A = austenite, BU = upper bainite;
 M = martensite
 N/A = not any

- Which corrosion mechanism is most serious in steels vs. chlorides?

(a) interstitial	☒ pitting	(c) sensitization	(d) galvanic
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- Which of the following slip systems belongs to BCC:

(a) (111)[1-2 -1]	(b) (1-10)[111]	(c) [111](1-10)	(d) (1-11)[11-1]
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- Guess the A_3 temperature of the alloy in the figure

(a) 850 °C	☒ 750 °C	(c) 950 °C	(d) 1050 °C
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at Point B
 $A_1 = 723^\circ\text{C}$
 $\gamma = \frac{0.4 - 0}{0.5} = 80\%$

- Guess the excess in temperature above A_3 for proper austenitization upon steel Normalizing

(a) 180 °C	(b) 30 °C	☒ 50 °C	(d) 80 °C
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- Which is NOT a feature of "steel austenitizing" process

(a) homogenization	☒ melting	(c) carbide dissolution	(d) atomic diffusion
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- For the alloy in the figure, estimate the approximate martensite fraction on quenching from B-temperature

(a) N/A	(b) ~0.1/0.5	☒ ~0.4/0.5	(d) ~100%
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- Give the Q-temperature value for the steel in TTT diagram (figure)

(a) A_2	(b) A_1	☒ A_1	(d) A_{cm}
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- Indicate the controlling mechanism of phase transformations in the R-field (figure)

(a) diffusion	☒ diffusionless	(c) no growth	(d) no transformation
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- TTT curves of steels shift with Si alloying to:

(a) right	☒ left	(c) right-down	(d) left-down
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- Which class of steels is strengthened by heat treatment

(a) austenitic stainless steels	(b) ferritic stainless steels	☒ tempering steels	(d) cold worked steels
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- Estimate the microstructure composition at point 2 in the TTT diagram (see figure)

a) 50%M+45%T	b) 50%M+50%AR	☒ 50%TM-BL+50%AR	d) 45%TM-BL+55%AR
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What is NOT a feature of pearlite

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|-----------------|------------------------|-----------------------------------|--|
| (a) it is phase | (b) it is a soft phase | (c) forms by diffusion on cooling | ☒ cannot be formed under CCT |
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- Which factor is not affected by hardenability in steels?

☒ toughness	(b) hardness	(c) martensite penetration	(d) amount of residual austenite
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- Guess the class of steel for X10CrNiTi1810

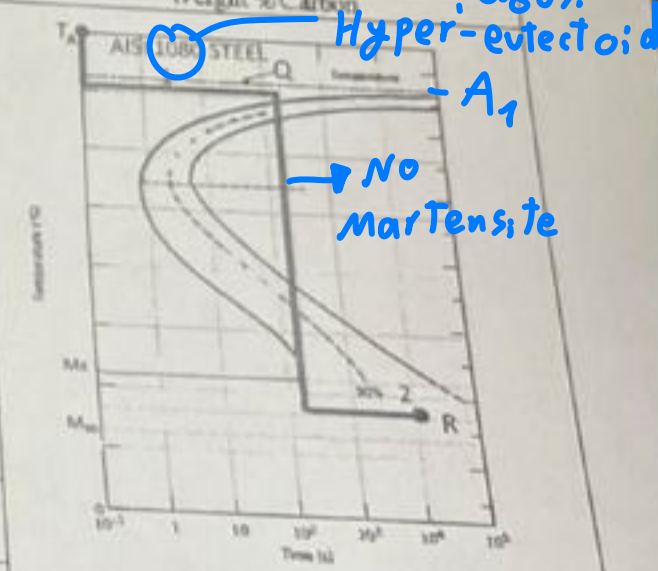
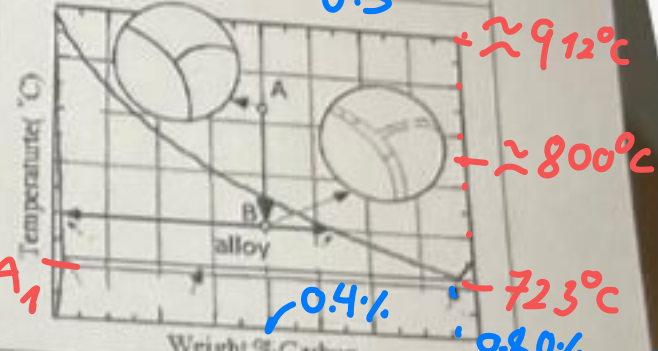
(a) tempered steel	☒ stainless steel	(c) low-alloy steel	(d) tool steel
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- In which surface treatment the compression stress state is largest?

(a) flame hardening	☒ laser hardening	(c) induction heat treatment	(d) carburizing
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- How ferritic stainless steels are strengthened to desired strength required for the application?

(a) heat treatment	(b) precipitation	(c) grain refinement	☒ cold work
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Graph is not clear

Not Sure

Not heat treatable

For Austenitic

WRITTEN EXAM OF TMM

1. It is FALSE on the advantages of pure metals vs. alloys that are: (a) > ductile (b) > corrosion resistant (c) tougher ☒ (d) stiffer				Legend of Symbols⁽¹⁾: TM = tempered martensite, B = bainite BL = lower bainite, P = pearlite AR = residual austenite, F = ferrite, A = austenite, BU = upper bainite; M = martensite SS = stainless steels T-line = temperature line + division operator / separator
2. It is FALSE in Humé-Rothery rules that all provide indications for compound formation depending on (a) Electronegativity difference ☒ (b) atom radii difference = 25% (c) valence (d) crystal structure				
3. What is formed when electronegativities of atoms differ? ☒ (a) intermetallic compound (b) liquid solution (c) solid solution (d) alloy steels				
4. Precipitation hardening does NOT involve the following step: (a) solutionizing (b) quenching (c) aging ☒ (d) phase change				
5. Determine %C in Y and the symbol of T-line in W (see figure) (a) 0.022%/Acm ☒ (b) 0.0025%/Acm (c) 0.0025%/A ₁ (d) 0.022%/A ₃				
6. How does A₁ change after alloying 5% Mn to initial Co-alloy ☒ (a) decreased (b) increased (c) unchanged (d) disappear				
7. For C₁-alloy in figure, which temperature must be reached under equilibrium before final quenching to form 5% cementite and 95% M (a) down to Tamb ☒ (b) down to Acm (c) down to A ₁ (d) down to A ₃				
8. Estimate the proportion of pearlite of Co-alloy at T₀ (a) ~10 % (b) ~40 % (c) ~0.2+7 % ☒ (d) 50 %				
9. Indicate the R-field name in the TTT diagram (figure) (a) F+P (b) F+cementite ☒ (c) stable A (d) BU				
10. Indicate the T-line in C-field name in the TTT diagram (figure) (a) A ₃ (b) A ₁ (c) Acm ☒ (d) A ₁				
11. How is microstructure toughness affected in the AISI 4140 steel with decreasing temperature in the TTT diagram (see figure) (a) increased ☒ (b) decreased (c) unaffected (d) undefined				
12. Which of the following alloys may contain some pro-eutectoid cementite upon slow cooling (a) AISI 1020 (b) AISI 4340 ☒ (c) AISI 1095 (d) AISI 304				
13. Estimate the proportion of cooling products on cooling down to point 2 (see figure) ☒ (a) 50% M + 50% TM-BL (b) 50% M + 45% TM-BL + 5% AR (c) 50% TM-BL + 50% AR (d) 90% TM-BL + 10% AR				
14. What is the more suited class of steels for induction hardening? (a) austenitic SS ☒ (b) tempering steel (c) low carbon steel (d) None				
15. In which alloy strain hardening is most effective for highest strength? (a) Aluminum alloys ☒ (b) Co-alloy (c) Carbon steels (d) Mg alloys				
16. How is the eutectic composed in the Al-Si alloy system? (a) Al + Si (b) α (solid solution) + β (solid solution) (c) Si+liquid ☒ (d) α (solid solution) + Si				
17. Which are the precipitates formed since early aging in A2XXX alloys? → 2 = copper ☒ (a) Al ₂ Cu GP zones (b) Al ₂ Mn GP zones (c) Al ₂ Mn ₃ GP zones (d) Al ₂ Mg GP zones				
18. In which metal the rate of corrosion is highest? (a) Cu (b) Fe ☒ (c) Zn (d) Ti				

IMPORTANT: TO AVOID CONFUSION WITH POSSIBLE CANCELLATIONS, FIRST USE A PENCIL OR ALTERNATIVELY, MARK WITH A DOT YOUR OPTION AND THEN FILL THE FINAL OPTION LETTER OF YOUR CHOICE WITH YOUR PEN. IN THE LIMIT A CROSS-MARKED (X) OPTION SIGNIFIES CANCELLATION.